COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Quiz # 3

Course Title:	VLSI Design	Course Code:	CPE434	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais	Programme Name:	BS Computer Engineering		
Semester:		Batch:		Date:	11st May 2026
Time Allowed:	**10 Minutes**		**Maximum Marks:**	**5**	
Student's Name			**Registration. No.:**		

Question # 1: **(PLO3-CLO3-C5)**

Design a circuit for the following function at transistor level with LEAn integration with Pass transistors:

$$F(A,B,C) = \sum(0,2,4,5,7)$$

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$$= A'B'C' + A'BC' + AB'C' + AB'C + ABC$$

$$= A'(1\cdot B'\cdot C' + 1\cdot B\cdot C' + 0\cdot B'\cdot C' + 0\cdot B'\cdot C + 0\cdot B\cdot C)$$
$$+ A(0\cdot B'\cdot C' + 0\cdot B\cdot C' + 1\cdot B'\cdot C' + 1\cdot B'\cdot C + 1\cdot B\cdot C)$$
$$= A'(B'C' + BC') + A(B'C' + B'C + BC)$$

$$B'C' + BC' = B'(1\cdot C' + 0\cdot C') + B(0\cdot C' + 1\cdot C')$$
$$= B'(C' + 0) + B(0 + C') = B'(C') + B(C')$$

$$B'C' + B'C + BC = B'(1\cdot C' + 1\cdot C + 0\cdot C) + B(0\cdot C' + 0\cdot C + 1\cdot C)$$
$$= B'(C' + C + 0) + B(0 + 0 + C)$$
$$= B'(C' + C) + B(0 + C)$$
$$= B'(1) + B(C)$$